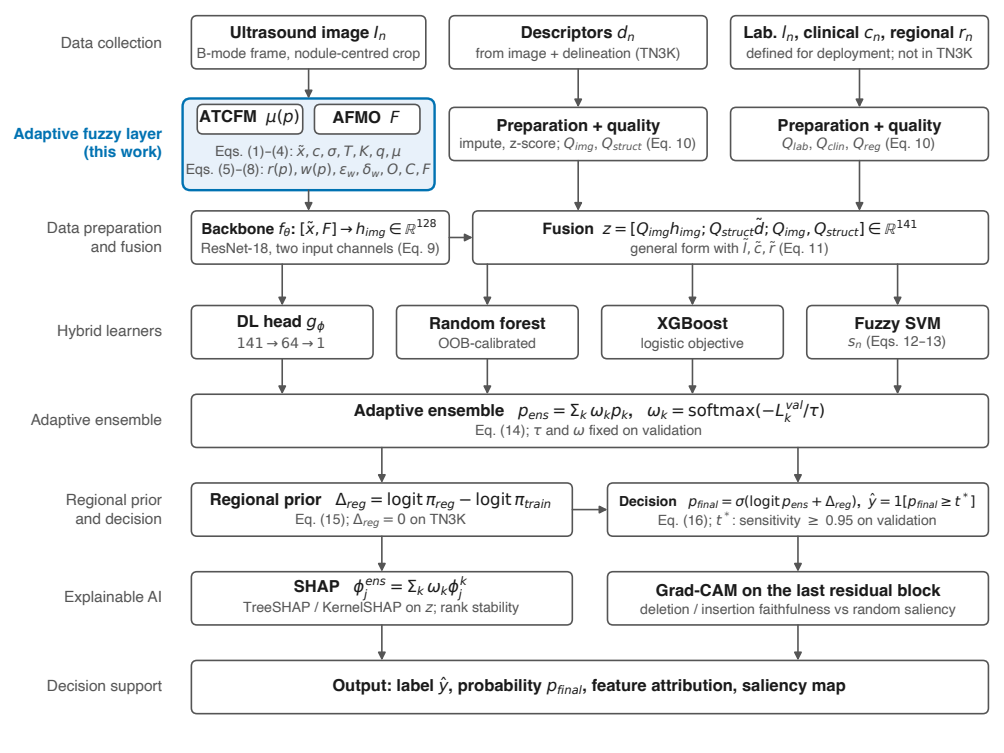

Data collection
Ultrasound image I_n
B-mode frame, nodule-centred crop
Descriptors d_n
from image + delineation (TN3K)
Lab. l_n, clinical c_n, regional r_n
defined for deployment; not in TN3K
Adaptive fuzzy layer (this work)
ATCFM $\mu(p)$
AFMO F
Eqs. (1)–(4): $\tilde{x}, c, \sigma, T, K, q, \mu$
Eqs. (5)–(8): $r(p), w(p), \varepsilon_w, \delta_w, O, C, F$
Preparation + quality
impute, z-score; Q_{img}, Q_{struct} (Eq. 10)
Preparation + quality
Q_{lab}, Q_{clin}, Q_{reg} (Eq. 10)
Data preparation and fusion
Backbone f_θ: $[\tilde{x}, F] \to h_{img} \in \mathbb{R}^{128}$
ResNet-18, two input channels (Eq. 9)
Fusion $z = [Q_{img} h_{img}; Q_{struct} \tilde{d}; Q_{img}, Q_{struct}] \in \mathbb{R}^{141}$
general form with $\tilde{l}, \tilde{c}, \tilde{r}$ (Eq. 11)
Hybrid learners
DL head g_ϕ
$141 \to 64 \to 1$
Random forest
OOB-calibrated
XGBoost
logistic objective
Fuzzy SVM
s_n (Eqs. 12–13)
Adaptive ensemble
Adaptive ensemble $p_{ens} = \Sigma_k \omega_k p_k, \quad \omega_k = \mathrm{softmax}(-L_k^{val}/\tau)$
Eq. (14); τ and ω fixed on validation
Regional prior and decision
Regional prior $\Delta_{reg} = \mathrm{logit}\,\pi_{reg} - \mathrm{logit}\,\pi_{train}$
Eq. (15); $\Delta_{reg} = 0$ on TN3K
Decision $p_{final} = \sigma(\mathrm{logit}\,p_{ens} + \Delta_{reg}), \ \hat{y} = 1[p_{final} \geq t^*]$
Eq. (16); t^*: sensitivity $\geq$ 0.95 on validation
Explainable AI
SHAP $\phi_j^{ens} = \Sigma_k \omega_k \phi_j^k$
TreeSHAP / KernelSHAP on z; rank stability
Grad-CAM on the last residual block
deletion / insertion faithfulness vs random saliency
Decision support
Output: label $\hat{y}$, probability p_{final}, feature attribution, saliency map